

ADR-008: Multi-Agent Orchestration

FR016 — sources/adr.md

Masterclass source package snapshot

Functional Requirement	FR-016
Edition	Engineering Learning Academy — Source Package snapshot
Status	Accepted (FR-016 M1–M4 close-out — **Complete / Frozen** ; binding
Audience	Academy generation and study (snapshot; SoT remains authoritative)
Document type	Source package snapshot (PDF)

Faithful rendering of a regenerable Masterclass source snapshot. Do not hand-edit; regenerate via python scripts/build_masterclass_package.py. Repository documentation remains the source of truth.

Contents

Historical note (M1)

ADR-008: Multi-Agent Orchestration

Status: Accepted (FR-016 M1–M4 close-out — **Complete / Frozen**; binding product posture **GO AS LEARNING PROOF ONLY**)

Date: 2026-08-06

Acceptance: [eval/fr016_multi_agent_orchestration.md](#)

M4: [eval/fr016_m4_evaluation.md](#)

Reaffirms: [ADR-003](#) (orchestration ≠ agent reasoning; no framework required without failing evidence); [ADR-005](#) (pipeline advances require owner authority); [ADR-006](#) (fail-closed truth; never waive); [ADR-007](#) (BOPA frozen; ToolPolicy sole admission for BOPA actions)

Does not amend: [ADR-001](#), [ADR-002](#), [ADR-004](#)

Spike: [eval/fr016_m0_engineering_spike.md](#) (Accepted with revisions)

Contracts: [eval/fr016_m1_orchestration_contracts.md](#)

Runtime / go-no-go: [eval/fr016_m2_supervisor_runtime.md](#)

Context

FR-015 delivered one bounded agent (BOPA) that coordinates post-acquisition package/truth readiness under deterministic ToolPolicy. OAT-001 confirmed BOPA is operationally ready. FR-016 asks whether multiple specialists under a supervisor add genuine value.

Owner-accepted M0 (with revisions) found:

- Splitting Prep / Truth / Review into separate agents that wrap the same services is **multi-agent theatre** and is rejected.
- Near-term commercial value is **modest**; FR-016 is approved primarily as a **constrained learning milestone** and as **architectural substrate** for future permission-separated capabilities (e.g. Job Discovery).
- The narrow topology is: Deterministic Orchestration Supervisor (DOS) + existing BOPA + read-only Operational Briefing Specialist (OBS) + typed handoffs + DelegationPolicy + per-specialist ToolPolicy.

Decision

1. **Introduce constrained multi-agent orchestration** under package `career_intelligence.multi_agent`. Do **not** place it in FR-008 `career_intelligence.orchestration`. Do **not** modify `career_intelligence.agent` (BOPA) unless a genuine integration defect is demonstrated.
2. **Topology (normative):** - **DOS** — deterministic supervisor; **delegates only**; performs no domain work; gains no specialist authority by sitting above specialists. - **BOPA** — existing FR-015 specialist; mutating allow-list **unchanged**. - **OBS** — new strictly read-only specialist that produces an `OperationalBrief` with cross-surface synthesis BOPA must not absorb by broadening

`prepare_for_owner_review`. - **Typed handoffs** — append-only, policy-validated, idempotent; never free-form chat; never grant another specialist's tools.

3. **Authority model:** - **DelegationPolicy** — sole admission authority for specialist invocation. - **Per-specialist ToolPolicy** — BOPA keeps `evaluate_action_policy`; OBS has `evaluate_obs_action_policy` over a read-only allow-list. - Global orchestration limits: max steps, max visits per specialist, repeated delegation, circular delegation, no-progress.
 4. **Decision policy:** Deterministic supervisor routing and deterministic specialist proposers are the **operational default**. Optional LLM may propose a specialist or brief wording **only** behind the same deterministic policies. LLM output never expands allow-lists.
 5. **Explicit theatre rejection:** Do **not** create Application Preparation, Truth, or Review "persona" agents that merely rename FR-010/011/014 services or BOPA actions. That pattern is multi-agent theatre.
 6. **Commercial posture:** Do **not** claim strong near-term commercial value for FR-016. Value claims are limited to learning outcomes and future extensibility until M2+ evidence shows otherwise.
 7. **Mandatory M2 go/no-go review** before M3/M4. At end of M2, evidence must answer: - Does DOS add value beyond direct `cic agent` / reporting commands? - Does OBS remove a meaningful owner task? - Do typed handoffs and separated permissions improve safety, auditability, or future extensibility? - Is the complexity proportionate? - Continue to M3/M4, keep as learning proof, or defer until Job Discovery?
 8. **Out of scope for FR-016:** job discovery / Seek / LinkedIn / Indeed / scraping; automatic submission; pipeline mutation; recruiter contact; truth waiver; agent-to-agent free-form conversation; role-play personas; framework adoption (LangGraph, OpenAI Agents SDK, Semantic Kernel / MAF, CrewAI) without a new ADR and failing product evidence under ADR-003 conditions.
 9. **Milestone delivery:** - **M1:** contracts + unit tests. - **M2:** DOS runtime, handoff store, OBS adapters (read-only), BOPA as child, audit — **go/no-go: GO AS LEARNING PROOF ONLY** ([eval/fr016_m2_supervisor_runtime.md](#)). - **M3–M4:** minimal close-out only (not product expansion) on owner request. M3 owner CLI complete ([eval/fr016_m3_owner_cli.md](#)); **M4 complete** — FR-016 frozen ([eval/fr016_m4_evaluation.md](#); [eval/fr016_multi_agent_orchestration.md](#)).
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Alternatives considered

Alternative	Why rejected / deferred
Keep BOPA only (defer FR-016)	Valid commercially; owner accepted constrained learning + substrate path
Prep / Truth / Review specialist cast	Multi-agent theatre; no distinct ToolPolicy value
LLM coordinator with chat handoffs	Weak deterministic control; injection surface; fashion risk
Hierarchical team	Unnecessary complexity for single-user CIC
Framework-hosted graph now	ADR-003 conditions unmet; custom contracts portable
Broaden BOPA instead of OBS	Would mix brief-only / batch-triage with mutating prepare goal

Consequences

- FR-016 is **Complete / Frozen / Accepted** as a learning proof ([eval/fr016_multi_agent_orchestration.md](#)).
- Binding product posture remains **GO AS LEARNING PROOF ONLY**; prefer `cic agent run` for ordinary preparation.
- BOPA behaviour remains frozen (ADR-007).
- Domain SoTs remain authoritative; orchestration/agent audits are additive.
- Future Acquisition specialist remains a **placeholder** only.
- FR-017 must not begin without explicit owner request.

Historical note (M1)

At M1 acceptance, contracts were frozen with no runtime until M2. M2–M4 subsequently delivered runtime, CLI, evaluation, and documentation freeze.

Guardrails

- DOS must not call mutating domain services directly.
- OBS must never gain `run_preparation`, `verify_package`, `validate_truth_package`, `submit`, `pipeline write`, or `truth-waive` tools.
- Handoffs must not widen ToolPolicy.
- Untrusted job/recruiter text must never become orchestration instructions.
- Do not reopen FR-008–FR-016 exit criteria without owner request.